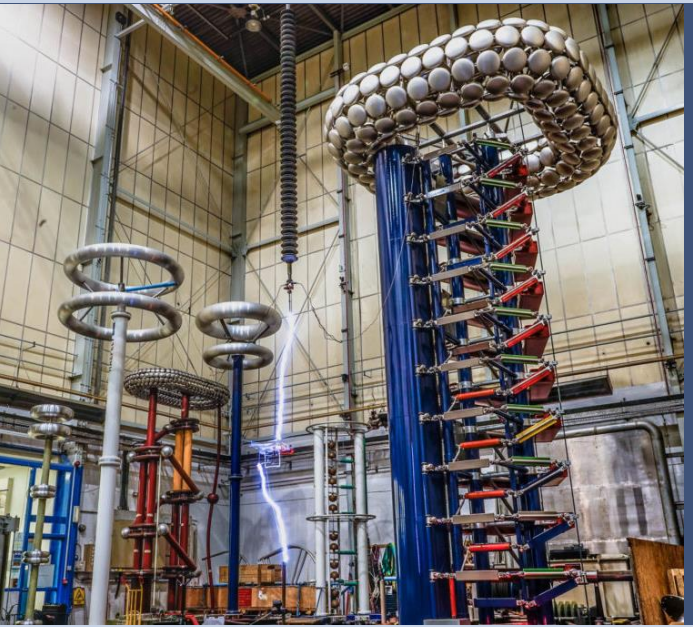


EE4047

High Voltage Techniques I

Week 1.1 – Monday Lesson

Asst. Prof. Dr. M. Alparslan Zehir

27.02.2023

Asst. Prof. Dr. M. Alparslan Zehir

E-mail: alparslan.zehir@marmara.edu.tr Room: M4-228



Education:

BSc, MSc and PhD: Istanbul Technical University (ITU), Dpt. of Electrical Engineering

Postdoctoral Researcher Experience:

2019-2020: University College Cork, MaREI (Marine and Renewable Energy Ireland) Centre, (Ireland)

Past Project Experience:

5 completed international projects (SEEMBEET, GReSBAS, GaMDER, DiNODR and CREDENCE)
5 ongoing national and international projects

My Courses in the Undergraduate Program:

Fall: EE2001 Electrical Circuits I **Spring:** EE4047 High Voltage Techniques I, EE110 Intr. to Elec. Eng., EE4049 Power System Protection

Areas of Research:

- **Power Systems**
- **Smart Grids**
- **Microgrids**
- **Cyber Security**
- **Renewable Energy**
- **Energy Storage**
- **Electric Vehicle Charging**
- **Virtual Power Plants**
- **Electricity Markets**
- **Power & Energy Management**
- **Building Automation Systems**

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Past experience specific to this course:

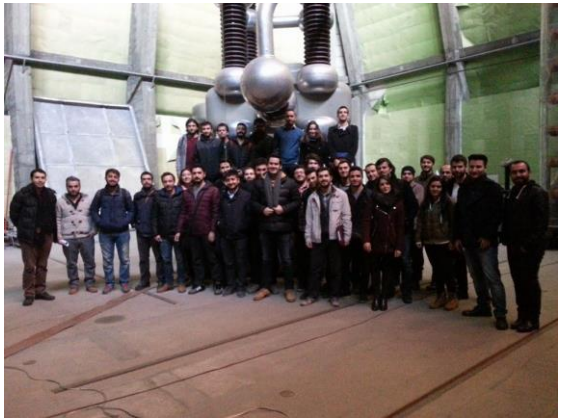
ITU Electrical Engineering Undergraduate program and Fuat Kulunk High Voltage Laboratory:

- High Voltage Laboratory Experiments
- High Voltage Techniques Course
- Discharge Events in High Voltage Technique Course
- Overvoltages and Insulation Coordination Course
- Industrial tests (up to 1.2 MV cont. And 3.6 MV impulse to TEIAS, SIEMENS and many other)

EPDK funded, over 1 million TL budget, R&D Project with BEDAS (Bogazici Elektrik Dagitim A.S.)









EE4047 – High Voltage Techniques I

INTRODUCTION

Turkish National Electric Power System and The Fundamentals of High Voltage Engineering

Week 1.1 – Monday Lesson

Asst. Prof. Dr. M. Alparslan Zehir

27.02.2023

OUTLINE

1 Introduction to the Course

2 Turkish National Electric Power System

3 The Fundamentals of High Voltage Engineering

1 INTRODUCTION TO THE COURSE

Objectives of the Course:

Introduce HV problems and HV applications, to evaluate discharge phenomena at high voltage systems, to introduce basic breakdown phenomena and their properties at gas, solid, and liquid dielectrics, introduce overvoltage and protection techniques.

Recommended Reading:

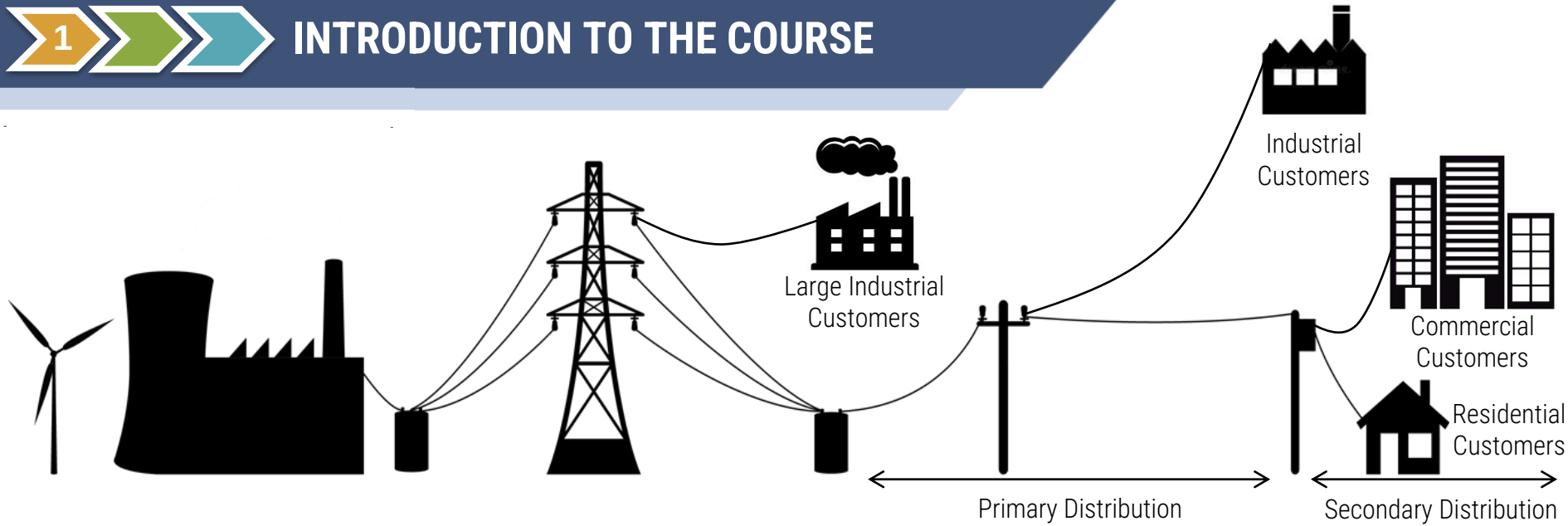
1. E. Kuffel, W. S. Zaengl, J. Kuffel, High Voltage Engineering Fundamentals, Pergamon Press, Oxford, 2000.
2. M. S. Naidu, V. Kamaraju, High Voltage Engineering, Tata McGraw-Hill, New Delhi, 2004.

Assessment:

1 midterm	:	40%
Homework or Quiz	:	20%
Final	:	40%

Minimum 70% attendance to online lessons is mandatory.

INTRODUCTION TO THE COURSE



Generation

(Centralized Power plants located near available resources)

10-30 kV

Transmission

(Transmit energy over long distances to load centers, using overhead lines and high voltage)

66-750 kV AC or
500-1500 kV DC

Distribution

(Transmit energy to its final destination using overhead lines, underground cables, using medium voltage and low voltage)
0.4 to 36 kV

Retail

(End-uses of customers such as lighting, appliances, heating etc.)

END OF THE LESSON

Any questions?

Check out the Power Engineering playlist with 32 videos on Youtube:

https://www.youtube.com/watch?v=JwRTpWZReJk&list=PLLZI7WaEmZi3F134eDujF5ZNGxA1E0dkm